Shreshth Srivastava

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ABOUT

Linux Kernel Developer at Intel, contributing to the **Linux x86 subsystem** (MSR interfaces, KVM/nested virtualization, FRED). Background spans **Android platform software** at Qualcomm and **CPU/GPU performance engineering** (ARM internship, CUDA kernel optimization). Proficient in **C, C++, and Python**.

WORK EXPERIENCE

Linux Kernel Developer *Intel, Hillsboro, OR — Jul 2026 – Present*

- Designed and built a three-level nested-**KVM** (L0→L1→L2) validation stack for Intel **FRED** (Flexible Return and Event Delivery), including a custom FRED-enabled kernel and QEMU; used **ftrace** to confirm correct FRED event-delivery behavior across all three virtualization layers.
- Took over a stalled x86 MSR header-consolidation series and delivered a **v3**, 5-patch revision submitted for upstream review, resolving naming objections raised by three separate reviewers.
- Root-caused an x86 NMI-source reporting bug by building a custom virtualization test environment and authoring a kernel selftest series to isolate and reproduce the failure.
- Contributed x86 kernel documentation fixes upstream, and independently verified correctness of an MSR interface series through binary-level regression analysis across the kernel build.

Network/ML Performance Engineer (Intern) *ARM, Austin, TX — May 2025 – Aug 2025*

- Profiled compute workloads using **Linux perf** and hardware performance counters (IPC/CPI, LLC misses, branch mispredictions, TLB pressure) to diagnose **CPU microarchitecture behavior** including pipeline stalls, cache contention, and memory bandwidth saturation across ARM64 and x86 platforms.
- Evaluated **compiler auto-vectorization** (ARM NEON / x86 AVX), `-O3/-march=native` build flags, and **thread affinity** to optimize parallel execution on multi-core systems; improved compute throughput by **15%** and packet processing by **12%**.
- Built **distributed benchmarking infrastructure** using Python, Terraform, and Ansible to deploy and validate compute-intensive workloads across heterogeneous ARM64 and x86 platforms, enabling reproducible regression and performance testing.
- Benchmarked across **NVIDIA Ampere Altra, Intel Xeon, AMD EPYC, and ARM Neoverse**; analyzed **per-platform memory bandwidth, pipeline efficiency**, and microarchitecture behavior to guide architecture-specific optimization.

Software Engineer *Qualcomm, India — Jul 2022 – Aug 2024*

- Cross-compiled and ported performance-critical **C++ applications** (OBS Studio, Chromium, GIMP) from x86 to ARM64 for Snapdragon targets, adapting build toolchains (Clang/GCC, CMake); tuned compiler flags reducing CPU utilization by 10–15%.
- Built Android and C# reference applications for **Qualcomm Enterprise SDK (QESDK)** APIs (Precise Positioning, CPU/GPU Boost, Low Latency Mode); debugged underlying Android/Linux services QESDK depends on (e.g., **modem service**) to support Low Latency Mode. Served as official integration reference for OEM partners including **Meta, Microsoft, Vivo, and Xiaomi**.
- Built a proof-of-concept peer-to-peer file-sharing application on Qualcomm's **Distributed Context Fabric (DCF)** for proximity-based device discovery; evaluated signal response time and detection latency for newly discovered devices.

Solution Delivery Analyst *Deloitte, India — Jul 2021 – May 2022*

- Built a **Java/Shell** server monitoring and automation framework for remote diagnostics across 80+ production Linux servers.
- Tuned **JVM** configurations (garbage collection, heap sizing) to reduce service response times, validating improvements via **AppDynamics**; reduced latency by **50%**.
- Migrated on-premises applications to **AWS**.
- Resolved user authentication issues integrating **PingID, SAML, and OAuth** across enterprise applications.

EDUCATION

University of Colorado Boulder *Aug 2024 – Apr 2026*

M.S. in Computer Science — GTA: Principles of Programming Languages (CSCI 3155)

Coursework: Computer Graphics, Design & Analysis of Algorithms, Network Systems, Neural Networks, Foundations of Software Engineering

Vellore Institute of Technology (VIT), India *Jun 2017 – Jun 2021*

B.Tech in Electronics and Communication Engineering — GPA: 8.58/10

Coursework: Data Structures & Algorithms, Operating Systems, Digital Logic Design, Microprocessors & Embedded Systems, Network Security

PROJECTS

CUDA GPU Kernel Performance Lab *CUDA C, C++, RTX 3070 Ti, Nsight Compute*

- Implemented and benchmarked CUDA matrix multiplication kernels on an **RTX 3070 Ti**: naive global memory baseline, **shared memory tiling**, register-level optimization, and loop unrolling; measured throughput gains at each stage against a **cuBLAS** baseline.
- Profiled kernels with **Nsight Compute** (SM occupancy, warp scheduling, memory coalescing, shared memory bank conflicts, L1/L2 hit rates) and roofline analysis to locate and fix memory-bound bottlenecks.

Encrypted Packet Filtering with eBPF *C, Linux Kernel, XDP/TC*

- Implemented **XDP/TC-based eBPF programs** that encrypt outgoing packet payloads in-kernel and decrypt them on ingress between two machines on the same LAN, so a passive observer running **tcpdump** on the shared network segment cannot recover the plaintext traffic.
- Built C++ user-space utilities to load the eBPF programs, manage keys, and retrieve telemetry via **BPF maps**; profiled kernel-space vs. user-space filtering/encryption overhead.

SKILLS

Languages: C, C++17/20, CUDA C, Python, Bash, Rust, Java

GPU & Parallel Programming: CUDA Programming Model, Thread/Block/Grid Hierarchy, Shared Memory, Memory Coalescing, Warp Scheduling, Nsight Compute, cuBLAS

Kernel & OS: Linux Kernel Development, KVM/Nested Virtualization, FRED, Xen, Kernel Debugging (GDB/kgdb/ftrace), Virtual Memory, Process Scheduling

Performance & Profiling: Linux perf, Hardware Performance Counters (IPC/CPI, LLC, TLB, NUMA), Microbenchmarking, SIMD (AVX/NEON), Compiler Optimization (`-O2/-O3/-march`), Roofline Model Analysis

Architecture: ARM64 (Snapdragon/Neoverse), x86 (Intel Xeon), GPU Architecture

Tooling: GDB, CMake, Git, QEMU, Terraform, Ansible, Docker